

MTE 203 – Advanced Calculus

Week 1 Homework

Vectors and 3D Analytic Geometry (S.11.1)¹

As mentioned in Lecture 1, the shortest distance between two points (or the length of the line segment joining two points) $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ in 3-D space is given by,

$$\|P_1P_2\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Using your previous knowledge on vector operations (MTE 119) and the concept of distance between points and lines in the 2-D space (MATH 118), solve the following problems in the 3-D space.

Problem 1 [S. 11.1, Prob. 5]:

Show that the (unidirected, perpendicular) distances from a point (x, y, z) to the x –, y –, and z – axes are, respectively, $\sqrt{y^2 + z^2}$, $\sqrt{x^2 + z^2}$, $\sqrt{y^2 + x^2}$.

Problem 2 [S. 11.1, Prob. 13]:

- If $(\sqrt{3} - 3, 2 + 2\sqrt{3}, 2\sqrt{3} - 1)$ and $(2\sqrt{3}, 4, \sqrt{3} - 2)$ are two vertices of an equilateral triangle, and if the third vertex lies on the z –axis, find the third vertex coordinates.
- Can you find a third vertex on the x –axis?

Planes and Lines (S.11.5)

Lines in 3D

Find the vector, parametric, and symmetric (if possible) equations for the following straight lines:

Problem 1 [S.11.5,13]:

Find the vector, parametric, and symmetric (if possible) equations for the straight line that passes through the point $(-1, 3, 6)$ and is parallel to the vector $(2, -3, 0)$

¹ This part of the homework is considered a review from past math courses as well as MTE 119 – Statics. It is material that is assumed you already know. For a refresher in vectors, please refer to the MTE 203 - Handout 1 and Sections 11.1 and 11.3 of your textbook.

Problem 2 [S.11.5,17]:

Find the vector, parametric, and symmetric (if possible) equations for the straight line that passes through the point $(1, -3, 5)$ and is parallel to the line

$$\frac{x}{5} = \frac{y-2}{3} = \frac{z+4}{-2}$$

Problem 3 [S.11.5,19]:

Find the vector, parametric, and symmetric (if possible) equations for the straight line that passes through the point of intersection of the lines,

$$\frac{x-1}{2} = \frac{y+4}{-3} = \frac{z-2}{5}$$

$$\frac{x-1}{2} = \frac{y+4}{3} = \frac{z-2}{4}$$

and is parallel to the line joining the points $(1, 3, -2)$ and $(2, -2, 1)$

Planes in 3D**Problem 1 [S.11.5, 9] :**

Find the equation of the plane containing the lines

$$3x + 4y = -6, x + 2y + z = 2$$

$$2y + 3z = 19, 3x - 2y - 9z = -58$$

Problem 2 [S.11.5, 11] :

The acute angle between two intersecting planes is defined as the acute angle between their normals. Find the acute angle between the planes,

$$x - 2y + 4z = 6$$

$$2x + y = z + 4$$

Problem 3 [S.11.5, 27] :

Verify that the equation of the plane passing through the three points $P_1(x_1, y_1, z_1)$, $P_2(x_2, y_2, z_2)$, and $P_3(x_3, y_3, z_3)$ can be written in the form,

$$\vec{P_1P} \cdot \vec{P_1P_2} \times \vec{P_1P_3} = 0$$

where $P(x, y, z)$ is any point in the plane.

Distance between Points, Lines and Planes (S.11.6)

Problem 1 [S.11.6, 21]:

Find the distance from the point $(1, 2, -3)$ to the line

$$x = 2(y + 1) = \frac{z - 4}{2}$$

Problem 2 [S.11.6, 25]:

Find the distance between the lines

$$\frac{x - 1}{2} = \frac{y + 3}{3} = 4 - z$$

and

$$x = -1 + t, \quad y = 2t, \quad z = 3 - 2t$$

Problem 3 [S. 11.6, 41]:

If a , b , and c (all positive constants) are the intercepts of a plane with the x -, y -, and z -axes, and p is the length of the perpendicular from the origin to the plane, show that

$$\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$$

Extra Practice Problems²

The answers to these problems, along with all even-numbered exercises, are provided at the back of the textbook. Detailed solutions can be found in Trim's Student Solution Manual. If you don't own a copy, a reserved edition is available at the Davis Centre.

1. S. 11.1, Probs. 4, 14, 20
2. S. 11.3, Probs. 8, 20, 30
3. S. 11.5, Probs. 4, 8, 18, 22
4. S. 11.6, Probs. 22, 26, 28, 40

² You are strongly encouraged to try and solve the problems assigned in this homework on your own, before looking at the solutions. These problems are only the minimum necessary to master the material. You are also encouraged to do additional problems from the text for practice (some suggested **extra practice problems** have been listed here).